Types of Inferential Statistics:

it consists of making some conclusions using some experiments based on collected data.

With inferential statistics, you are trying to reach conclusions that extend beyond the immediate data alone. For instance, we use inferential statistics to try to infer from the sample data what the population might think.

1) Hypothesis Testing:
2) Regression Analysis:

Hypothesis Testing:

Hypothesis testing is a type of inferential statistics that is used to test assumptions and draw conclusions about the population from the available sample data.

It involves setting up a **null hypothesis** and an **alternative hypothesis** followed by conducting a statistical test of significance. A conclusion is drawn based on the value of the test statistic, the critical value, and the confidence intervals.

A hypothesis test can be **left-tailed, right-tailed, or two-tailed.**

Null Value Testing/Null Hypothesis:

This is the default or initial assumption. It often represents a statement of no effect, no difference, or no change. It is denoted as H0.

Null hypothesis testing is a formal approach to decide whether a statistical relationship in a sample reflects a real relationship in the population or is just due to chance.

Alternative hypothesis:

This is the statement you want to test or the claim you're investigating. It represents the opposite of the null hypothesis and is what you hope to find evidence for. It is denoted as Ha or sometimes as H1.

The alternative hypothesis is a statement used in statistical inference experiments. **It is contradictory to the null hypothesis.**

Points to be noted for Hypothesis Testing:

If the output value **is within** the Confidence Interval then we can say the **Null hypothesis is True and Accepted**. In this scenario, we can say we fail to reject the Alternate Hypothesis.

If the output value **is not within** the Confidence Interval then we can say **Null HyPothis is Rejcted** and **Alternate Hypothesis gets Accepted.**

Errors in Hypothesis Testing

Type 1 error :

it happens when the null hypothesis of an experiment is true but rejected often called a false positive.

type 2 error:

it occurs when the null hypothesis is false but still not rejected, also known as a false negative.

Which error is considered to be more dangerous?

Type 1 error is considered to be worse or more dangerous than type 2 because to reject what is true is more harmful than keeping the data that is not true.

Different hypothesis tests are used in Hypothesis Testing.

- Z Test:
- T-Test:
- F Test:
- Annona Test:
- Chi-Square Test: